Exercises — Semi-logarithmic graphs and extrapolation

3A — 30/10/2026

Newton's law of cooling: $T - T_{room} = (T_0 - T_{room})\, e^{-kt}$, *so* $\ln(T - T_{room}) = \ln(T_0 - T_{room}) - k\,t$.

Exercise 1 — Which graph is a straight line?

A potato cools in a room at $T_{\text{room}} = 21\,^{\circ}\text{C}$. Its temperature is measured every 5 minutes.

t / min	0	5	10	15	20
T / °C	61.0	49.3	41.0	35.1	31.0

(a) Complete a table of $T - T_{\text{room}}$ and $\ln(T - T_{\text{room}})$ (3 decimal places).

(b) Is the graph of T against t a straight line? Is the graph of $\ln(T - T_{\text{room}})$ against t a straight line? Explain using the values.

(c) Find the gradient of the second graph and the value of k in min^{-1}.

Solution — (a) $T - T_{\text{room}}$: 40.0, 28.3, 20.0, 14.1, 10.0 K. ln: $\ln 40.0 = 3.689$, $\ln 28.3 = 3.343$, $\ln 20.0 = 2.996$, $\ln 14.1 = 2.646$, $\ln 10.0 = 2.303$.

(b) The differences $T - T_{\text{room}}$ fall by different amounts each step (-11.7, -8.3, -5.9, -4.1), so T against t is a curve. The values of ln fall by almost the same amount each step (-0.346, -0.347, -0.350, -0.343), so the second graph is a straight line.

$$\text{gradient} = \frac{\Delta y}{\Delta t} = \frac{2.303 - 3.689}{20 - 0} = -0.0693\ \text{min}^{-1}$$

$$k = -\text{gradient} = 0.0693\ \text{min}^{-1}$$

Check: the difference halves every 10 min ($40 \to 20 \to 10$), and $t_{1/2} = \ln 2/k = 0.693/0.0693 = 10$ min.

Exercise 2 — Gradient of the best-fit line

A student plots $y = \ln(T - T_{\text{room}})$ against t (in s) for a cooling potato and draws the best-fit line. Two points read *on the line*, far apart, are (200 s, 3.05) and (1000 s, 2.65). $T_{\text{room}} = 20\,^{\circ}\text{C}$.

(a) Calculate the gradient and k, with units.

(b) Calculate the intercept at $t = 0$ and the temperature T_0 of the potato at Start.

(c) Calculate the time $1/k$ and the half-life of the difference $T - T_{\text{room}}$.

(d) Why must the points of the gradient triangle be taken *on the line*, far apart, and not

from the table of data?

Solution —

(a) $\text{gradient} = \dfrac{2.65 - 3.05}{1000 - 200} = -5.0 \times 10^{-4}\ \text{s}^{-1}, \qquad k = 5.0 \times 10^{-4}\ \text{s}^{-1}$

(b) $y(0) = 3.05 - (-5.0 \times 10^{-4})(200) = 3.05 + 0.10 = 3.15$

$T_0 - T_{\text{room}} = e^{3.15} = 23.3\ \text{K} \ \Rightarrow\ T_0 = 43.3\,^\circ\text{C}$

(c) $1/k = 2000\ \text{s} \approx 33\ \text{min}, \qquad t_{1/2} = \dfrac{\ln 2}{k} = \dfrac{0.693}{5.0 \times 10^{-4}} = 1.39 \times 10^3\ \text{s} \approx 23\ \text{min}$

(d) A single data point carries its own random error; a point on the best-fit line averages them. Points far apart (more than half the line) make Δy and Δt large, so the same reading error is a small fraction of them: the gradient is more precise.

Exercise 3 — Extending the line back

A potato leaves a water bath at $T_{\text{bath}} = 45\,^\circ\text{C}$ and is left in a room at $T_{\text{room}} = 22\,^\circ\text{C}$. The best-fit line of $y = \ln(T - T_{\text{room}})$ against t is

$$y = 2.90 - 3.0 \times 10^{-4}\, t \qquad (t \text{ in s},\ t = 0 \text{ at Start}).$$

(a) Calculate $\ln(T_{\text{bath}} - T_{\text{room}})$.

(b) Find the time at which the line reaches this value. How many minutes before Start did the potato leave the bath?

(c) Check that your answer is sensible: calculate the temperature of the potato at Start. Is it below T_{bath}?

(d) Would you expect the real time to be longer or shorter than your estimate? (Think of how the centre of the potato cools at the beginning.)

Solution —

(a) $\ln(45 - 22) = \ln 23 = 3.135$

(b) $3.135 = 2.90 - 3.0 \times 10^{-4}\, t^* \ \Rightarrow\ t^* = \dfrac{3.135 - 2.90}{-3.0 \times 10^{-4}} = -7.8 \times 10^2\ \text{s} \approx -13\ \text{min}$

The potato left the bath about 13 minutes before Start.

(c) $T(0) = 22 + e^{2.90} = 22 + 18.2 = 40.2\,^\circ\text{C} < 45\,^\circ\text{C}$. Consistent: it was cooling for 13 minutes before Start. (If $T(0)$ were above T_{bath} there would be an error.)

(d) Longer. Just after leaving the bath the centre is still near T_{bath}, so the true curve is flatter at the start than the straight line; the straight line reaches the bath level too late (closer to Start), and the estimate is short. The size of this systematic error depends on the size of the potato and the depth of the sensor.

Exercise 4 — Two traps

(a) A student uses the data of Exercise 1 but plots $\ln T$ against t instead of $\ln(T - T_{\text{room}})$. Calculate the gradient between $t = 0$ and $t = 20$ min and compare it with the value of Exercise 1. What error would this make in k and in the half-life?

(b) The Arduino thermometer reads $+1.0\,^{\circ}\text{C}$ too high everywhere (zero error). The student reads both T and T_{room} with it. $T_{\text{bath}} = 50\,^{\circ}\text{C}$ is measured with a correct reference thermometer; the true room temperature is $22\,^{\circ}\text{C}$, the gradient is -3.0×10^{-4} s^{-1}. Does the zero error change the gradient? And the intercept? How does it change $\ln(T_{\text{bath}} - T_{\text{room}})$ and the estimated time?

Solution — (a) $\ln 61.0 = 4.111$ and $\ln 31.0 = 3.434$:

$$\text{gradient} = \frac{3.434 - 4.111}{20} = -0.0339 \text{ min}^{-1} \quad \text{(Exercise 1: } -0.0693\text{)}.$$

The k obtained is about half the correct one (0.034 against 0.069 min^{-1}) and the half-life is about 20 min instead of 10 min. $\ln T$ is not linear in t: the decay is of $T - T_{\text{room}}$, not of T.

(b) Every reading is $1.0\,^{\circ}\text{C}$ high, including T_{room}, so $T - T_{\text{room}}$ is unchanged: **gradient and intercept do not change**. But T_{bath} comes from another thermometer: the student calculates $T_{\text{bath}} - T_{\text{room}} = 50 - 23 = 27$ K instead of 28 K.

$$\ln 27 - \ln 28 = 3.296 - 3.332 = -0.036$$

$$\Delta t = \frac{-0.036}{-3.0 \times 10^{-4}} \approx 120 \text{ s} = 2 \text{ min}$$

The meeting point moves 2 min towards Start: the estimate becomes 2 min shorter. A zero error cancels in a difference only if *both* temperatures are read with the same instrument.